SkillUp

Project Documentation Template

Skill-Building Project: Performing Data Analytics to Find Trends Through Visualization

About This Template

This template is intended to help you document your project activities in a structured manner. As you progress through the tasks in the project, you should continue updating this document. By the end, you will have a complete and well-organized submission ready for expert evaluation.

Naming convention

You can save the document as **Project Documentation - [Your Initials].**

How to use the template

For each task and the steps within it, take screenshots of your code and any resulting output. Paste these into the appropriate placeholders provided in the document.

In certain tasks, you will also need to record your observations, such as trends you notice in the data. Placeholders have been included for these notes as well.

Important guidelines

You can follow these guidelines:

- Keep the document neatly formatted and easy to read.
- Use clear and professional language.
- Remember, your documentation should demonstrate not just your technical skills, but also on how effectively you:
    - Summarize key findings from the project,
    - Highlight the business implications of your analysis, and
    - Present your work in a professional and organized manner

Best of luck with your project!

Task 0: Framing the project

Step number and instruction	Area for screenshot, code, and observation
Step 1 Briefly describe the problem this project is addressing. Use your own words to summarize the issue being solved.	The problem addressed in this project is understanding the factors that influence Airbnb listing prices, availability, and host performance. Airbnb operates globally and has thousands of listings with different characteristics such as location, room type, pricing, and host ratings. Without proper analysis, it can be difficult to identify patterns in pricing, understand what makes certain listings more successful than others, and determine how service factors affect bookings.
Step 2 State the main outcome you aim to achieve through this project.	The main objective of this project is to analyze Airbnb listing data to identify pricing trends, location-based variations, and factors that influence host performance. Through data cleaning, exploratory data analysis, and visualization, the goal is to provide insights that can help optimize pricing strategies and improve service quality for hosts and the platform.
Step 3 Describe the dataset, its source, and the type of data it contains.	The dataset used in this project is the Airbnb Open Data dataset obtained from Kaggle. It contains information about Airbnb property listings including host details, neighborhood location, room type, price, availability, service fees, review scores, cancellation policies, and house rules. The data is provided in CSV format and allows analysts to explore trends related to pricing, location, host performance, and customer feedback.
Step 4 Explain who is affected by this problem and why the solution matters.	The stakeholders affected by this analysis include Airbnb hosts, guests, and the Airbnb platform itself. Hosts benefit by understanding how pricing and property characteristics affect bookings. Guests benefit from improved service quality and fair pricing strategies. Airbnb benefits by using data-driven insights to improve marketplace efficiency and customer satisfaction.

Step 5 List the tools, libraries, and technologies you plan to use.	The tools and technologies used in this project include: • Python for data analysis • Pandas for data manipulation • NumPy for numerical operations • Matplotlib and Seaborn for data visualization • VS Code for coding environment • Kaggle dataset for Airbnb listing data

Task 1: Loading the dataset

Step number and instruction	Area for screenshot, code, and observation

Step 1
Read the CSV file and load it into a pandas DataFrame.
	import pandas as pd
	import numpy as np
	import matplotlib.pyplot as plt
	import seaborn as sns
	
	# Load dataset
	df = pd.read_csv("Airbnb_Open_Data.csv")

Step 2
Display the first five rows of the dataset to understand the data format.

```
df.head()
```

[2] ✓ 0.1s Python

	id	NAME	host id	host_identity_verified	host name	neighbourhood group	neighbourhood	lat	long	country	...	service fee	minimum nights	number of reviews	last review	reviews per month	review rate number	calculat ho listin cou
0	1001254	Clean & quiet apt home by the park	80014485718	unconfirmed	Madaline	Brooklyn	Kensington	40.64749	-73.97237	United States	...	$193	10.0	9.0	10/19/2021	0.21	4.0	
1	1002102	Skylit Midtown Castle	52335172823	verified	Jenna	Manhattan	Midtown	40.75362	-73.98377	United States	...	$28	30.0	45.0	5/21/2022	0.38	4.0	
2	1002403	THE VILLAGE OF HARLEM....NEW YORK !	78829239556	NaN	Elise	Manhattan	Harlem	40.80902	-73.94190	United States	...	$124	3.0	0.0	NaN	NaN	5.0	
3	1002755	NaN	85098326012	unconfirmed	Garry	Brooklyn	Clinton Hill	40.68514	-73.95976	United States	...	$74	30.0	270.0	7/5/2019	4.64	4.0	
4	1003689	Entire Apt: Spacious Studio/Loft by central park	92037596077	verified	Lyndon	Manhattan	East Harlem	40.79851	-73.94399	United States	...	$41	10.0	9.0	11/19/2018	0.10	3.0	

5 rows × 26 columns

Step 3
Display the data types of each column to understand the nature of each variable and identify any potential issues with data types.

```
df.dtypes
```

[3] ✓ 0.0s

```
id                                int64
NAME                             object
host id                           int64
host_identity_verified           object
host name                        object
neighbourhood group              object
neighbourhood                    object
lat                             float64
long                            float64
country                          object
country code                     object
instant_bookable                 object
cancellation_policy              object
room type                        object
Construction year               float64
price                            object
service fee                      object
minimum nights                  float64
number of reviews               float64
last review                      object
reviews per month               float64
review rate number              float64
calculated host listings count  float64
availability 365                float64
house_rules                      object
license                          object
dtype: object
```

Task 2: Removing unnecessary fields

Step number and instruction	Area for screenshot, code, and observation
Step 1 Remove the following unwanted columns from the dataset: *host id*, *id*, *country*, and *country code*. Be sure to include screenshots of the dataset before and after eliminating these columns.	

```python
df_before = df.copy()

df = df.drop(columns=['host id', 'id', 'country', 'country code'])

df.head()
```

[4] ✓ 0.2s Python

	NAME	host_identity_verified	host name	neighbourhood group	neighbourhood	lat	long	instant_bookable	cancellation_policy	room type	...	service fee	minimum nights	number of reviews	last review	reviews per month
0	Clean & quiet apt home by the park	unconfirmed	Madaline	Brooklyn	Kensington	40.64749	-73.97237	False	strict	Private room	...	$193	10.0	9.0	10/19/2021	0.21
1	Skylit Midtown Castle	verified	Jenna	Manhattan	Midtown	40.75362	-73.98377	False	moderate	Entire home/apt	...	$28	30.0	45.0	5/21/2022	0.38
2	THE VILLAGE OF HARLEM....NEW YORK !	NaN	Elise	Manhattan	Harlem	40.80902	-73.94190	True	flexible	Private room	...	$124	3.0	0.0	NaN	NaN
3	NaN	unconfirmed	Garry	Brooklyn	Clinton Hill	40.68514	-73.95976	True	moderate	Entire home/apt	...	$74	30.0	270.0	7/5/2019	4.64
4	Entire Apt: Spacious Studio/Loft by central park	verified	Lyndon	Manhattan	East Harlem	40.79851	-73.94399	False	moderate	Entire home/apt	...	$41	10.0	9.0	11/19/2018	0.10

5 rows × 22 columns

Step 2 Document the reason for excluding these columns from your data analysis. This will demonstrate your ability to distinguish between relevant and irrelevant data.	The columns host id, id, country, and country code were removed because they do not contribute to the pricing or service analysis. These identifiers are useful for database indexing but do not provide meaningful insights for trend analysis, visualization or pricing strategy evaluation.

Task 3: Handling missing values and duplicate records

Step number and instruction	Area for screenshot, code, and observation
Step 1 Identify missing values in the dataframe and display their count in ascending order. If any values are missing, impute them according to the data type of the columns.	

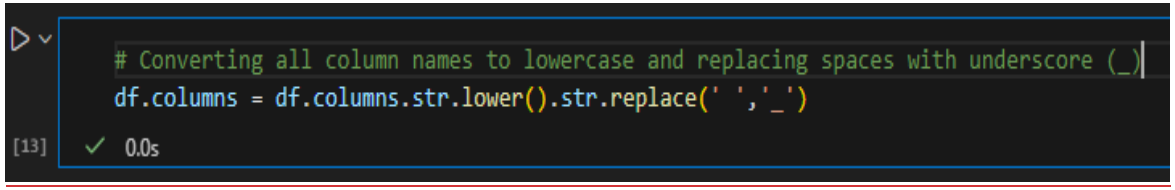
```python
df.isnull().sum().sort_values()
```

[5] ✓ 0.0s

```
room type                          0
long                               8
lat                                8
neighbourhood                     16
neighbourhood group               29
cancellation_policy               76
instant_bookable                 105
number of reviews                183
Construction year                214
price                            247
NAME                             250
service fee                      273
host_identity_verified           289
calculated host listings count   319
review rate number               326
host name                        406
minimum nights                   409
availability 365                 448
reviews per month              15879
last review                    15893
house_rules                    52131
license                       102597
dtype: int64
```

Step 2 Check whether there are any duplicate values in the dataframe and if present, remove them.	``` df.duplicated().sum() [15] ✓ 0.2s ... np.int64(3461) rows_before = df.shape[0] df = df.drop_duplicates() rows_after = df.shape[0] [16] ✓ 0.3s ```
Step 3 Display the total number of records in the dataframe before and after removing the duplicates to confirm that data cleaning has been effectively performed.	``` print("Records before removing duplicates: ", rows_before) print("Records after removing duplicates: ",rows_after) [17] ✓ 0.0s ... Records before removing duplicates: 102599 Records after removing duplicates: 99138 ```

Task 4: Transforming data

Step number and instruction	Area for screenshot, code, and observation
Step 1 Rename the column *availability 365* to *days_booked* to make the column name more intuitive and business-friendly.	
Step 2 Convert all column names to lowercase and replace the spaces in the column names with an underscore symbol.	
Step 3 Remove the dollar sign and comma from the columns *price* and *service_fee*.	

If necessary, convert these two columns to the appropriate data type. This is to ensure that financial data can be analyzed correctly.	[16] ✓ 0.0s
	`# Removing the dollar sign ($) and comma (,) from the columns price and service_fee.`
	`df['price'] = df['price'].replace(r'[\$,]', '', regex=True).astype(float)`
	`df['service_fee'] = df['service_fee'].replace(r'[\$,]', '', regex=True).astype(float)`

Task 5: Exploring the data

Step number and instruction	Area for screenshot, code, and observation

Step 1
Display the count of various room types available in the dataset.
	[17] ✓ 0.0s
	`# Showing the count of various room types available in the dataset.`
	`df['room_type'].value_counts()`
	…
	`room_type`
	`Entire home/apt  51987`
	`Private room  44887`
	`Shared room  2149`
	`Hotel room  115`
	`Name: count, dtype: int64`

Step 2

Display the room type with the strictest cancellation policy.

Describe the risk factors for both hosts and guests.

```
df.groupby('room_type')['cancellation_policy'].value_counts()
```

[19] ✓ 0.1s

```
room_type        cancellation_policy
Entire home/apt  moderate              17389
                 flexible              17360
                 strict                17238
Hotel room       flexible                 44
                 moderate                 37
                 strict                   34
Private room     moderate              15121
                 strict                14936
                 flexible              14830
Shared room      strict                  718
                 moderate                717
                 flexible                714
Name: count, dtype: int64
```

Listings with stricter cancellation policies may reduce risk for hosts by ensuring commitment from guests, but they may discourage bookings from guests who prefer flexible travel options.

Step 3

Display the average price per neighborhood group.
 	
Mention the most expensive neighborhood for rentals.
	``python
	# Showing the average price per neighborhood group
	avg_price = df.groupby('neighbourhood_group')['price'].mean().sort_values(ascending=False)
	avg_price
	``
	[20] ✓ 0.0s
	
	``
	neighbourhood_group
	Queens 628.668822
	Brooklyn 625.471627
	Bronx 625.271511
	Staten Island 625.060870
	Manhattan 621.666140
	brookln 580.000000
	manhatan 460.000000
	Name: price, dtype: float64
	``
	
	The neighborhood with the highest average rental price is **Queens**, indicating it is a premium area likely influenced by demand, location advantages, or property quality.

Task 6: Visualizing data – Part 1

Step number and instruction	Area for screenshot, code, and observation
Step 1 Create a horizontal bar chart to display	

the top 10 most expensive neighborhoods in the dataset.

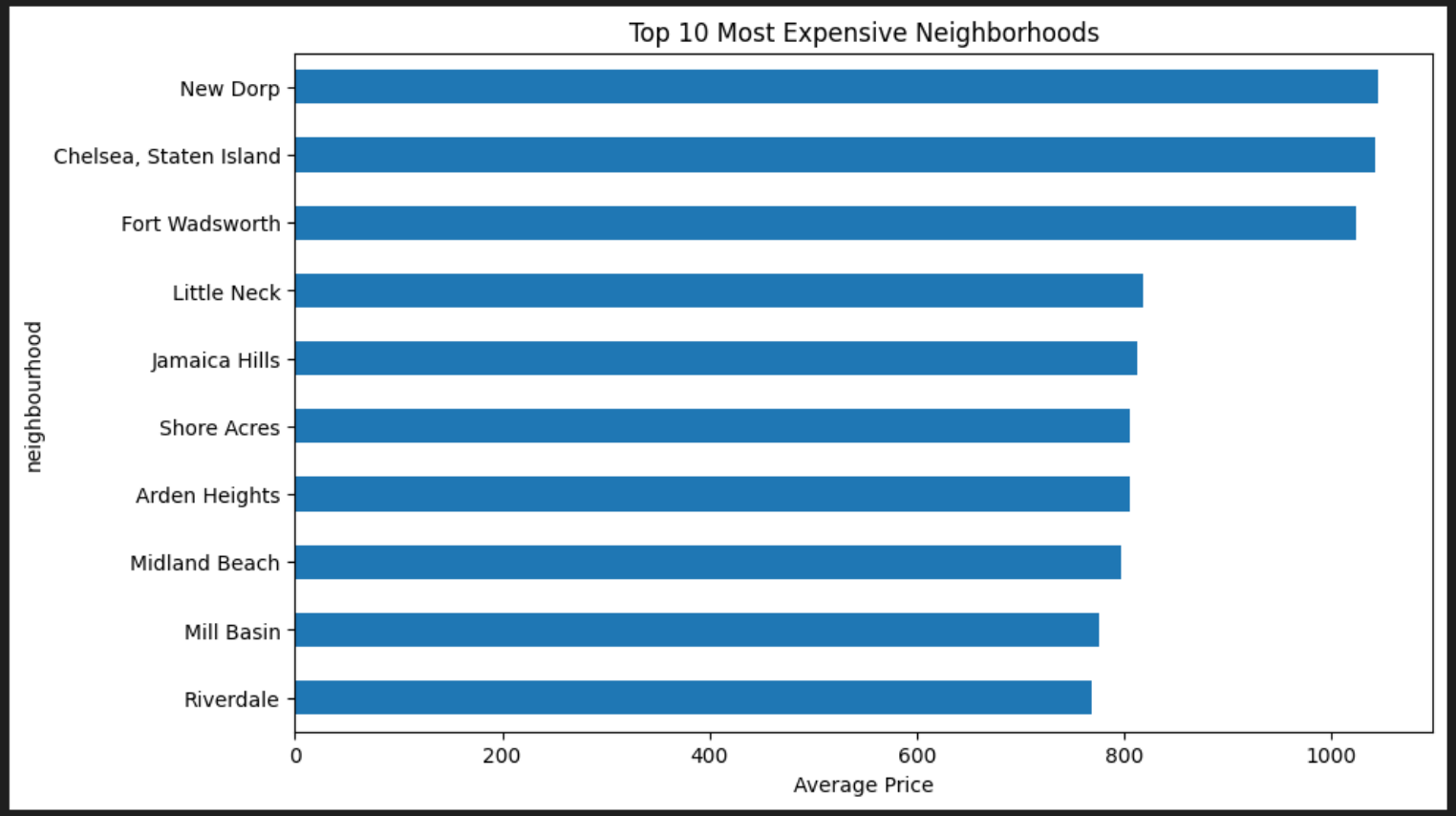
Create another chart with the 10 least expensive neighborhoods in the dataset.

```python
# Top 10 expensive neighborhoods

top10 = df.groupby('neighbourhood')['price'].mean().sort_values(ascending=False).head(10)

plt.figure(figsize=(10,6))
top10.sort_values().plot(kind='barh')
plt.title("Top 10 Most Expensive Neighborhoods")
plt.xlabel("Average Price")
plt.show()
```

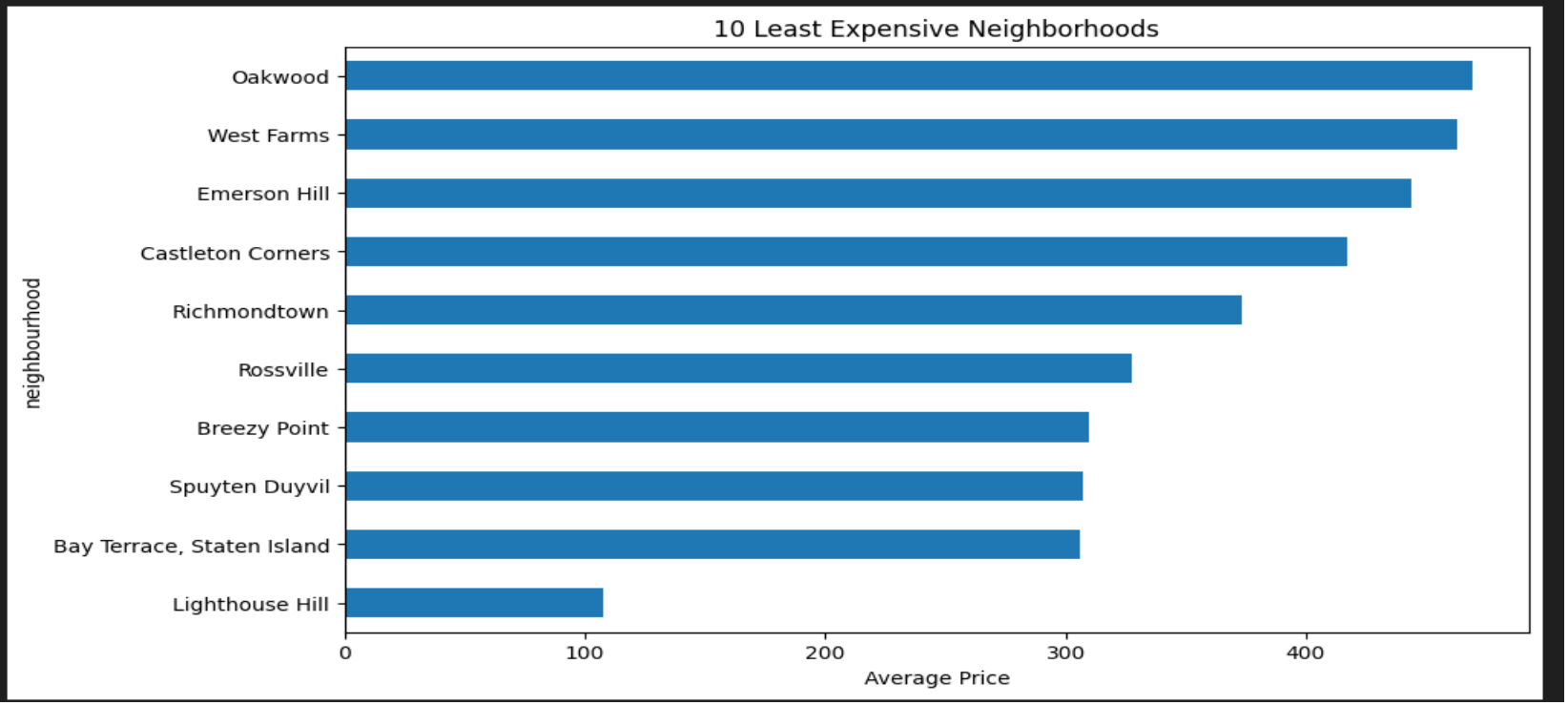
[21] ✓ 3.0s



```python
# Cheapest neighborhoods
cheap10 = df.groupby('neighbourhood')['price'].mean().sort_values().head(10)

plt.figure(figsize=(10,6))
cheap10.plot(kind='barh')
plt.title("10 Least Expensive Neighborhoods")
plt.xlabel("Average Price")
plt.show()
```

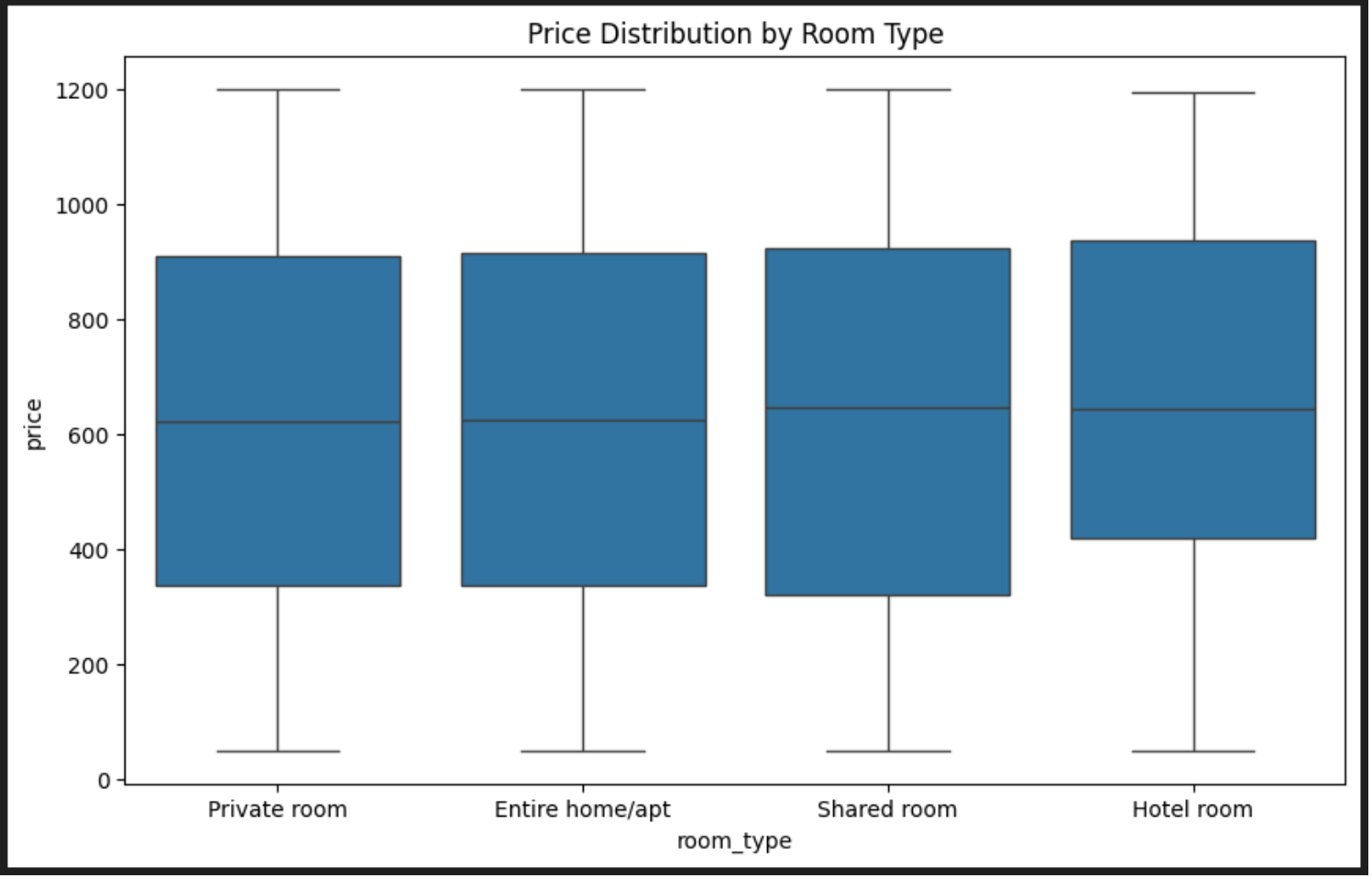
[22] ✓ 0.2s



Step 2 Create a box and whisker chart that showcases the price distribution of all listings split by room type to understand how prices vary within different accommodation categories.	

```python
# Boxplot price by room type
plt.figure(figsize=(10,6))
sns.boxplot(data=df, x='room_type', y='price')
plt.title("Price Distribution by Room Type")
plt.show()
```

[23] ✓ 0.5s



Task 7: Visualizing data – Part 2

Step number and instruction	Area for screenshot, code, and observation
Step 1 Create a scatter plot to illustrate the relationship between the cleaning fee and room price. Note any correlation and comment whether cleaning fees influence pricing strategies.	

```python
# Scatter plot cleaning fee vs Price
plt.figure(figsize=(8,6))
sns.scatterplot(data=df, x='service_fee', y='price')
plt.title("Cleaning Fee vs Price")
plt.show()
```

[26] ✓ 0.6s

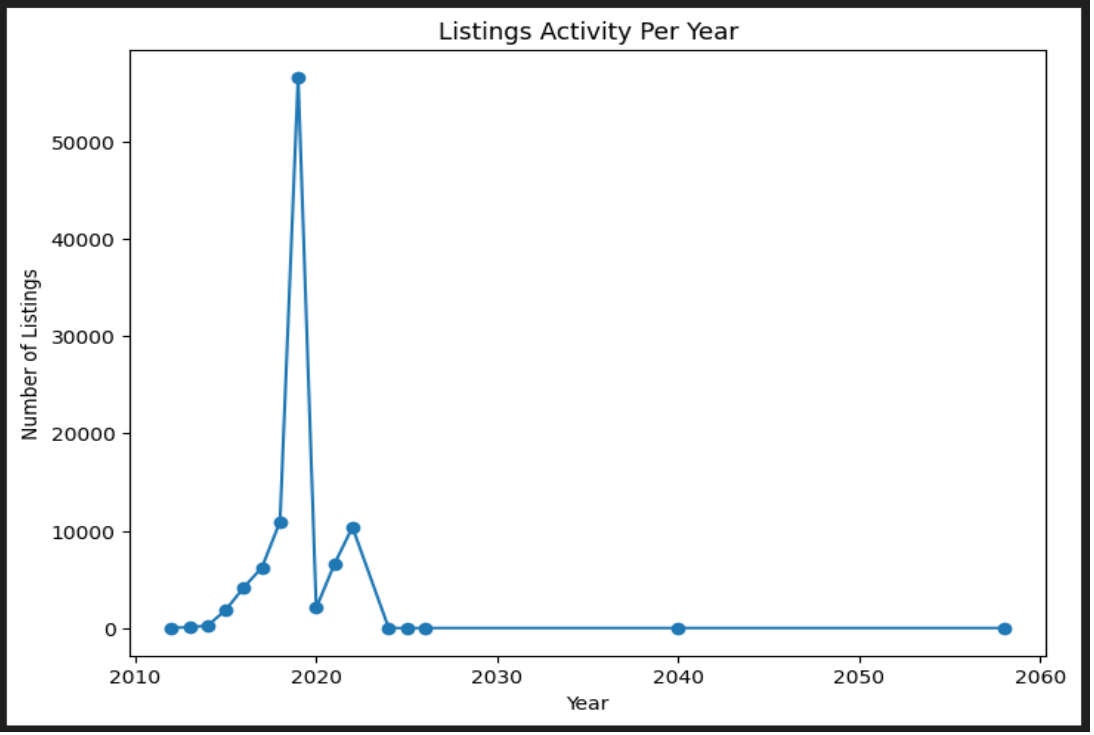
	The analysis reveals a positive relationship between listing price and service fee, indicating that higher-priced properties typically include higher service costs. However, the presence of distinct clusters suggests that some listings follow predefined pricing structures rather than strictly proportional pricing. This pattern may reflect standardized fee policies applied by hosts or the platform.
Step 2 Create a line chart to showcase the total amount of listings available per year.	

```python
# Line Charts - Listings per year
df['last_review'] = pd.to_datetime(df['last_review'])

listings_year = df['last_review'].dt.year.value_counts().sort_index()

plt.figure(figsize=(8,6))
listings_year.plot(kind='line', marker='o')
plt.title("Listings Activity Per Year")
plt.xlabel("Year")
plt.ylabel("Number of Listings")
plt.show()
```

[27] ✓ 0.4s

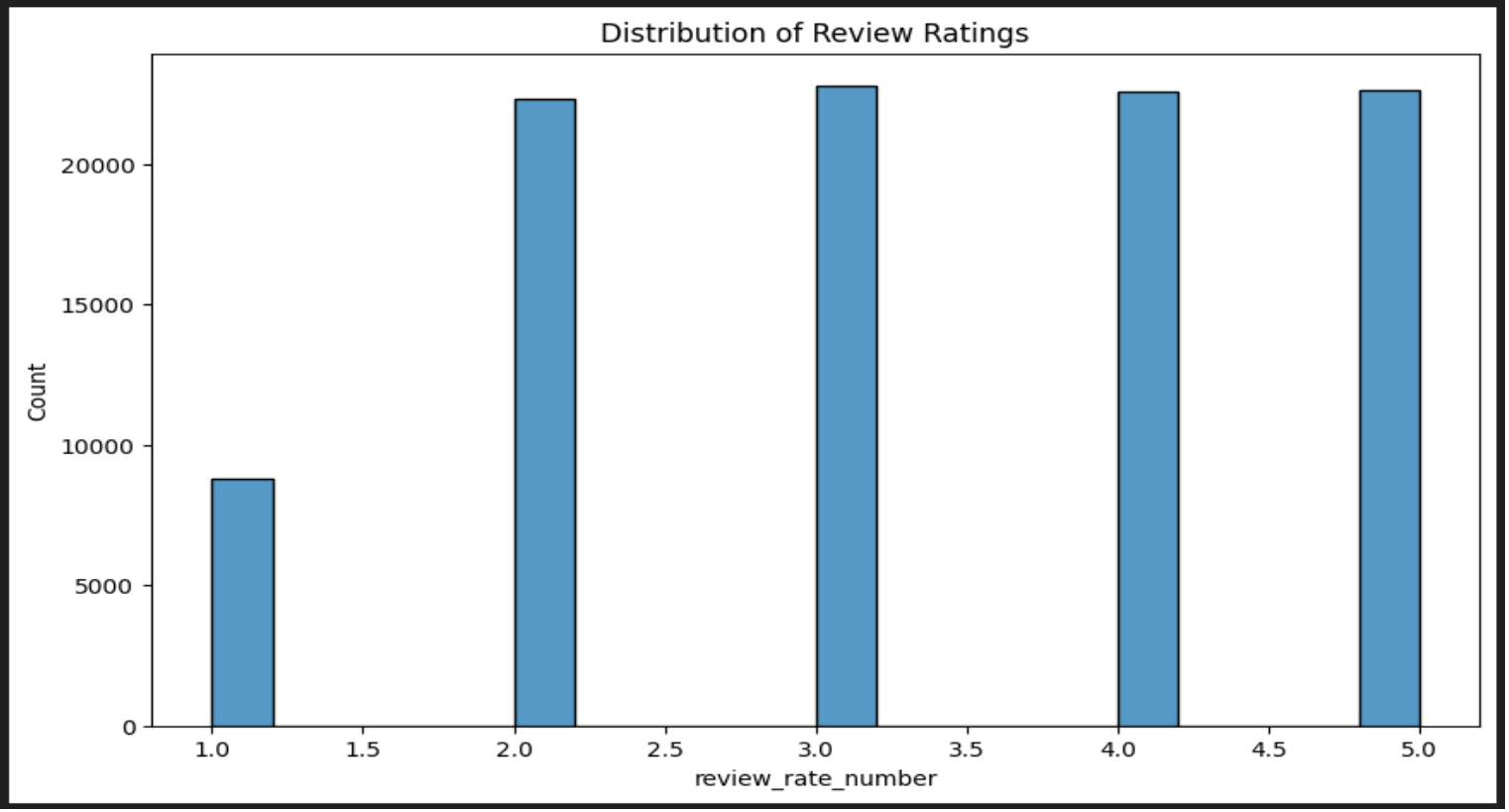


Task 8: Visualizing data – Part 3

Step number and instruction	Area for screenshot, code, and observation
Step 1 Create a data visualization of your choice using one of the review columns in isolation or combination with another column.	

```python
# Visualization using review scores
plt.figure(figsize=(10,6))
sns.histplot(df['review_rate_number'], bins=20)
plt.title("Distribution of Review Ratings")
plt.show()
```

[28] ✓ 0.3s



Step 2
Create a visualization to compare at least two different variables between super hosts and regular hosts.

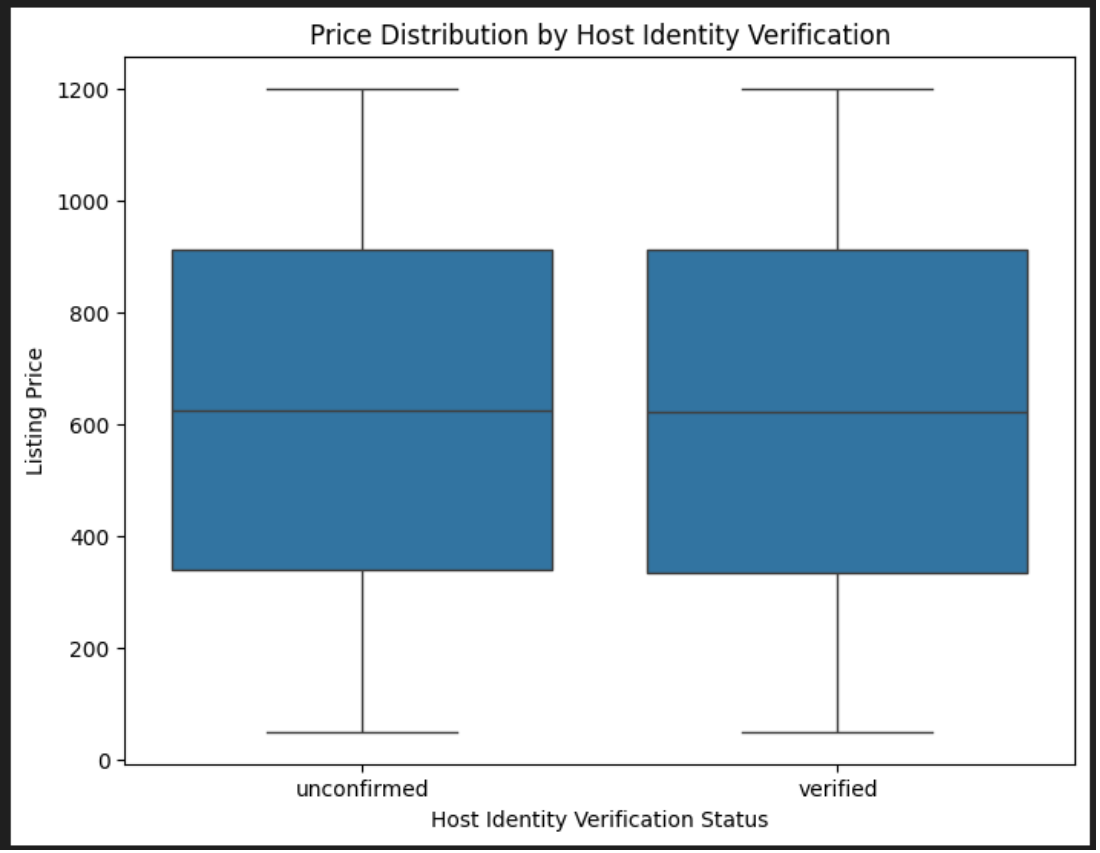
Identify and document the performance differences between highly-rated and average hosts.

```
# Comparing Superhosts vs Regular hosts
plt.figure(figsize=(8,6))
sns.boxplot(data=df, x='host_identity_verified', y='price')

plt.title("Price Distribution by Host Identity Verification")
plt.xlabel("Host Identity Verification Status")
plt.ylabel("Listing Price")

plt.show()
```

[30] ✓ 0.6s



	The analysis comparing verified and unverified hosts shows that listing prices are distributed similarly across both groups. The median prices and interquartile ranges appear nearly identical, suggesting that identity verification alone does not significantly influence pricing strategies. This indicates that other factors such as location, property type, and amenities may play a stronger role in determining listing prices.